

```
In [1]:
```

```
import pandas as pd
```

```
In [2]:
```

```
df = pd.read_csv(r'Sample.csv')
```

```
df
```

	Row	Order	Order	Ship	Ship	Customer	Customer		Segment	Country/Region		Postal	Re
	ID	ID	Date	Date	Mode	ID	Name				City...	Code	
0	1	US- 2019- 103800	03- 01- 2019	07- 01- 2019	Standard Class	DP-13000	Darren Powers		Consumer	United States	Houston	...77095	Ce
1	2	US- 2019- 112326	04- 01- 2019	08- 01- 2019	Standard Class	PO-19195	Phillina Ober		Home Office	United States	Naperville	...60540	Ce
2	3	US- 2019- 112326	04- 01- 2019	08- 01- 2019	Standard Class	PO-19195	Phillina Ober		Home Office	United States	Naperville	...60540	Ce
3	4	US- 2019- 112326	04- 01- 2019	08- 01- 2019	Standard Class	PO-19195	Phillina Ober		Home Office	United States	Naperville	...60540	Ce
4	5	US- 2019- 141817	05- 01- 2019	12- 01- 2019	Standard Class	MB- 18085	Mick Brown		Consumer	United States	Philadelphia	...19143	Ea
...	...	...	...	...	...	...	...	...	...	...	...	...	...
10189	10190	US- 2022- 143259	30- 12- 2022	03- 01- 2023	Standard Class	PO-18865	Patrick O'Donnell		Consumer	United States	New York City	...10009	Ea
10190	10191	US- 2022- 115427	30- 12- 2022	03- 01- 2023	Standard Class	EB-13975	Erica Bern		Corporate	United States	Fairfield	...94533	We
10191	10192	US- 2022- 156720	30- 12- 2022	03- 01- 2023	Standard Class	JM-15580	Jill Matthias		Consumer	United States	Loveland	...80538	We
10192	10193	US- 2022- 143259	30- 12- 2022	03- 01- 2023	Standard Class	PO-18865	Patrick O'Donnell		Consumer	United States	New York City	...10009	Ea
10193	10194	CA- 2022- 143500	30- 12- 2022	03- 01- 2023	Standard Class	HO- 15230	Harry Olson		Consumer	Canada	Charlottetown...	...C0A	Ea
10194 rows × 21 columns													

	Row ID	Order ID	Order Date	Ship Date	Ship Mode \
0	1	US-2019-103800	03-01-2019	07-01-2019	Standard Class
1	2	US-2019-112326	04-01-2019	08-01-2019	Standard Class
2	3	US-2019-112326	04-01-2019	08-01-2019	Standard Class
3	4	US-2019-112326	04-01-2019	08-01-2019	Standard Class
4	5	US-2019-141817	05-01-2019	12-01-2019	Standard Class
...	...	...	...	...	...
10189	10190	US-2022-143259	30-12-2022	03-01-2023	Standard Class
10190	10191	US-2022-115427	30-12-2022	03-01-2023	Standard Class
10191	10192	US-2022-156720	30-12-2022	03-01-2023	Standard Class
10192	10193	US-2022-143259	30-12-2022	03-01-2023	Standard Class
10193	10194	CA-2022-143500	30-12-2022	03-01-2023	Standard Class

	Customer ID	Customer Name	Segment	Country/Region \
0	DP-13000	Darren Powers	Consumer	United States
1	PO-19195	Phillina Ober	Home Office	United States
2	PO-19195	Phillina Ober	Home Office	United States
3	PO-19195	Phillina Ober	Home Office	United States
4	MB-18085	Mick Brown	Consumer	United States
...	...	...	...	...
10189	PO-18865	Patrick O'Donnell	Consumer	United States
10190	EB-13975	Erica Bern	Corporate	United States
10191	JM-15580	Jill Matthias	Consumer	United States
10192	PO-18865	Patrick O'Donnell	Consumer	United States
10193	HO-15230	Harry Olson	Consumer	Canada

	City	Postal Code	Region	Product ID \
0	Houston	77095	Central	OFF-PA-10000174
1	Naperville	60540	Central	OFF-BI-10004094
2	Naperville	60540	Central	OFF-LA-10003223
3	Naperville	60540	Central	OFF-ST-10002743
4	Philadelphia	19143	East	OFF-AR-10003478
...	...	...	...	...
10189	New York City	10009	East	OFF-BI-10003684
10190	Fairfield	94533	West	OFF-BI-10004632
10191	Loveland	80538	West	OFF-FA-10003472
10192	New York City	10009	East	TEC-PH-10004774
10193	Charlottetown	C0A	East	OFF-BI-10004040

	Category	Sub-Category \
0	Office Supplies	Paper
1	Office Supplies	Binders
2	Office Supplies	Labels
3	Office Supplies	Storage
4	Office Supplies	Art
...	...	...
10189	Office Supplies	Binders
10190	Office Supplies	Binders
10191	Office Supplies	Fasteners
10192	Technology	Phones
10193	Office Supplies	Binders

	Product Name	Sales	Quantity \
0	Message Book, Wirebound, Four 5 1/2" X 4" Form...	16.448	2
1	GBC Standard Plastic Binding Systems Combs	3.540	2
2	Avery 508	11.784	3
3	SAFCO Boltless Steel Shelving	272.736	3
4	Avery Hi-Liter EverBold Pen Style Fluorescent ...	19.536	3
...	...	...	...
10189	Wilson Jones Legal Size Ring Binders	52.776	3
10190	GBC Binding covers	20.720	2
10191	Bagged Rubber Bands	3.024	3
10192	Gear Head AU3700S Headset	90.930	7
10193	Wilson Jones Impact Binders	3.024	3

	Discount	Profit
0	0.2	5.5512
1	0.8	-5.4870
2	0.2	4.2717

```
3      0.2 -64.7748
4      0.2  4.8840
...     ...     ...
10189   0.2 19.7910
10190   0.2  6.4750
10191   0.2 -0.6048
10192   0.0  2.7279
10193   0.2 -0.6048
```

```
[10194 rows x 21 columns]
```

```
In [9]:
```

```
# 1. Load the Sample Superstore dataset and display the first and last 10 rows.
```

```
print("First 10 rows:")
print(df.head(10))
```

```
print("\nLast 10 rows:")
print(df.tail(10))
```

stdout:

First 10 rows:

Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	\
0	1 US-2019-103800	03-01-2019	07-01-2019	Standard Class	DP-13000	
1	2 US-2019-112326	04-01-2019	08-01-2019	Standard Class	PO-19195	
2	3 US-2019-112326	04-01-2019	08-01-2019	Standard Class	PO-19195	
3	4 US-2019-112326	04-01-2019	08-01-2019	Standard Class	PO-19195	
4	5 US-2019-141817	05-01-2019	12-01-2019	Standard Class	MB-18085	
5	6 US-2019-167199	06-01-2019	10-01-2019	Standard Class	ME-17320	
6	7 US-2019-167199	06-01-2019	10-01-2019	Standard Class	ME-17320	
7	8 US-2019-106054	06-01-2019	07-01-2019	First Class	JO-15145	
8	9 US-2019-167199	06-01-2019	10-01-2019	Standard Class	ME-17320	
9	10 US-2019-167199	06-01-2019	10-01-2019	Standard Class	ME-17320	

Customer Name	Segment	Country/Region	City	Postal Code	\
0 Darren Powers	Consumer	United States	Houston	77095	
1 Phillina Ober	Home Office	United States	Naperville	60540	
2 Phillina Ober	Home Office	United States	Naperville	60540	
3 Phillina Ober	Home Office	United States	Naperville	60540	
4 Mick Brown	Consumer	United States	Philadelphia	19143	
5 Maria Etezadi	Home Office	United States	Henderson	42420	
6 Maria Etezadi	Home Office	United States	Henderson	42420	
7 Jack O'Briant	Corporate	United States	Athens	30605	
8 Maria Etezadi	Home Office	United States	Henderson	42420	
9 Maria Etezadi	Home Office	United States	Henderson	42420	

Region	Product ID	Category	Sub-Category	\
0 Central	OFF-PA-10000174	Office Supplies	Paper	
1 Central	OFF-BI-10004094	Office Supplies	Binders	
2 Central	OFF-LA-10003223	Office Supplies	Labels	
3 Central	OFF-ST-10002743	Office Supplies	Storage	
4 East	OFF-AR-10003478	Office Supplies	Art	
5 South	FUR-CH-10004063	Furniture	Chairs	
6 South	OFF-AR-10001662	Office Supplies	Art	
7 South	OFF-AR-10002399	Office Supplies	Art	
8 South	OFF-BI-10004632	Office Supplies	Binders	
9 South	OFF-FA-10001883	Office Supplies	Fasteners	

	Product Name	Sales	Quantity	\
0	Message Book, Wirebound, Four 5 1/2" X 4" Form...	16.448	2	
1	GBC Standard Plastic Binding Systems Combs	3.540	2	
2	Avery 508	11.784	3	
3	SAFCO Boltless Steel Shelving	272.736	3	
4	Avery Hi-Liter EverBold Pen Style Fluorescent ...	19.536	3	
5	Global Deluxe High-Back Manager's Chair	2573.820	9	
6	Rogers Handheld Barrel Pencil Sharpener	5.480	2	
7	Dixon Prang Watercolor Pencils, 10-Color Set w...	12.780	3	
8	Ibico Hi-Tech Manual Binding System	609.980	2	
9	Alliance Super-Size Bands, Assorted Sizes	31.120	4	

Discount	Profit
0 0.2	5.5512
1 0.8	-5.4870
2 0.2	4.2717
3 0.2	-64.7748
4 0.2	4.8840
5 0.0	746.4078
6 0.0	1.4796
7 0.0	5.2398
8 0.0	274.4910
9 0.0	0.3112

[10 rows x 21 columns]

Last 10 rows:

Row ID	Order ID	Order Date	Ship Date	Ship Mode	\
10184	10185 CA-2022-141531	29-12-2022	02-01-2023	Standard Class	
10185	10186 CA-2022-141531	29-12-2022	02-01-2023	Standard Class	
10186	10187 US-2022-143259	30-12-2022	03-01-2023	Standard Class	
10187	10188 US-2022-126221	30-12-2022	05-01-2023	Standard Class	
10188	10189 US-2022-115427	30-12-2022	03-01-2023	Standard Class	

10189	10190	US-2022-143259	30-12-2022	03-01-2023	Standard Class
10190	10191	US-2022-115427	30-12-2022	03-01-2023	Standard Class
10191	10192	US-2022-156720	30-12-2022	03-01-2023	Standard Class
10192	10193	US-2022-143259	30-12-2022	03-01-2023	Standard Class
10193	10194	CA-2022-143500	30-12-2022	03-01-2023	Standard Class

	Customer ID	Customer Name	Segment	Country/Region	\
10184	BG-18435	Bruce Galang	Consumer	Canada	
10185	BG-18435	Bruce Galang	Consumer	Canada	
10186	PO-18865	Patrick O'Donnell	Consumer	United States	
10187	CC-12430	Chuck Clark	Home Office	United States	
10188	EB-13975	Erica Bern	Corporate	United States	
10189	PO-18865	Patrick O'Donnell	Consumer	United States	
10190	EB-13975	Erica Bern	Corporate	United States	
10191	JM-15580	Jill Matthias	Consumer	United States	
10192	PO-18865	Patrick O'Donnell	Consumer	United States	
10193	HO-15230	Harry Olson	Consumer	Canada	

	City	...	Postal Code	Region	Product ID	\
10184	Quebec City	...	G1B	East	FUR-FU-10004093	
10185	Quebec City	...	G1B	East	OFF-FA-10000089	
10186	New York City	...	10009	East	FUR-BO-10003441	
10187	Columbus	...	47201	Central	OFF-AP-10002457	
10188	Fairfield	...	94533	West	OFF-BI-10002103	
10189	New York City	...	10009	East	OFF-BI-10003684	
10190	Fairfield	...	94533	West	OFF-BI-10004632	
10191	Loveland	...	80538	West	OFF-FA-10003472	
10192	New York City	...	10009	East	TEC-PH-10004774	
10193	Charlottetown	...	C0A	East	OFF-BI-10004040	

	Category	Sub-Category	\
10184	Furniture	Furnishings	
10185	Office Supplies	Fasteners	
10186	Furniture	Bookcases	
10187	Office Supplies	Appliances	
10188	Office Supplies	Binders	
10189	Office Supplies	Binders	
10190	Office Supplies	Binders	
10191	Office Supplies	Fasteners	
10192	Technology	Phones	
10193	Office Supplies	Binders	

	Product Name	Sales	Quantity	\
10184	Hand-Finished Solid Wood Document Frame	68.460	2	
10185	Acco Glide Clips	19.600	5	
10186	Bush Westfield Collection Bookcases, Fully Ass...	323.136	4	
10187	Eureka The Boss Plus 12-Amp Hard Box Upright V...	209.300	2	
10188	Cardinal Slant-D Ring Binder, Heavy Gauge Vinyl	13.904	2	
10189	Wilson Jones Legal Size Ring Binders	52.776	3	
10190	GBC Binding covers	20.720	2	
10191	Bagged Rubber Bands	3.024	3	
10192	Gear Head AU3700S Headset	90.930	7	
10193	Wilson Jones Impact Binders	3.024	3	

	Discount	Profit
10184	0.0	20.5380
10185	0.0	9.6040
10186	0.2	12.1176
10187	0.0	56.5110
10188	0.2	4.5188
10189	0.2	19.7910
10190	0.2	6.4750
10191	0.2	-0.6048
10192	0.0	2.7279
10193	0.2	-0.6048

[10 rows x 21 columns]

```
In [16]:  
  
# 2. Identify and handle missing values using at least two different techniques.  
  
df.isnull().sum()  
  
# Technique 1 - Imputation (Fill Missing Values)  
  
df['Sales'] = df['Sales'].fillna(df['Sales'].mean())  
  
df['Category'] = df['Category'].fillna(df['Category'].mode()[0])  
  
print(df)  
  
# Technique 2: Deletion (Drop Missing Values)  
  
df_cleaned = df.dropna()
```

stdout:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode \
0	1	US-2019-103800	03-01-2019	07-01-2019	Standard Class
1	2	US-2019-112326	04-01-2019	08-01-2019	Standard Class
2	3	US-2019-112326	04-01-2019	08-01-2019	Standard Class
3	4	US-2019-112326	04-01-2019	08-01-2019	Standard Class
4	5	US-2019-141817	05-01-2019	12-01-2019	Standard Class
...	...	...	...	...	...
10189	10190	US-2022-143259	30-12-2022	03-01-2023	Standard Class
10190	10191	US-2022-115427	30-12-2022	03-01-2023	Standard Class
10191	10192	US-2022-156720	30-12-2022	03-01-2023	Standard Class
10192	10193	US-2022-143259	30-12-2022	03-01-2023	Standard Class
10193	10194	CA-2022-143500	30-12-2022	03-01-2023	Standard Class

	Customer ID	Customer Name	Segment	Country/Region \
0	DP-13000	Darren Powers	Consumer	United States
1	PO-19195	Phillina Ober	Home Office	United States
2	PO-19195	Phillina Ober	Home Office	United States
3	PO-19195	Phillina Ober	Home Office	United States
4	MB-18085	Mick Brown	Consumer	United States
...	...	...	...	...
10189	PO-18865	Patrick O'Donnell	Consumer	United States
10190	EB-13975	Erica Bern	Corporate	United States
10191	JM-15580	Jill Matthias	Consumer	United States
10192	PO-18865	Patrick O'Donnell	Consumer	United States
10193	HO-15230	Harry Olson	Consumer	Canada

	City	Postal Code	Region	Product ID \
0	Houston	77095	Central	OFF-PA-10000174
1	Naperville	60540	Central	OFF-BI-10004094
2	Naperville	60540	Central	OFF-LA-10003223
3	Naperville	60540	Central	OFF-ST-10002743
4	Philadelphia	19143	East	OFF-AR-10003478
...	...	...	...	...
10189	New York City	10009	East	OFF-BI-10003684
10190	Fairfield	94533	West	OFF-BI-10004632
10191	Loveland	80538	West	OFF-FA-10003472
10192	New York City	10009	East	TEC-PH-10004774
10193	Charlottetown	C0A	East	OFF-BI-10004040

	Category	Sub-Category \
0	Office Supplies	Paper
1	Office Supplies	Binders
2	Office Supplies	Labels
3	Office Supplies	Storage
4	Office Supplies	Art
...	...	...
10189	Office Supplies	Binders
10190	Office Supplies	Binders
10191	Office Supplies	Fasteners
10192	Technology	Phones
10193	Office Supplies	Binders

	Product Name	Sales	Quantity \
0	Message Book, Wirebound, Four 5 1/2" X 4" Form...	16.448	2
1	GBC Standard Plastic Binding Systems Combs	3.540	2
2	Avery 508	11.784	3
3	SAFCO Boltless Steel Shelving	272.736	3
4	Avery Hi-Liter EverBold Pen Style Fluorescent ...	19.536	3
...	...	...	...
10189	Wilson Jones Legal Size Ring Binders	52.776	3
10190	GBC Binding covers	20.720	2
10191	Bagged Rubber Bands	3.024	3
10192	Gear Head AU3700S Headset	90.930	7
10193	Wilson Jones Impact Binders	3.024	3

	Discount	Profit
0	0.2	5.5512
1	0.8	-5.4870
2	0.2	4.2717
3	0.2	-64.7748

```

4          0.2    4.8840
...          ...      ...
10189      0.2   19.7910
10190      0.2    6.4750
10191      0.2   -0.6048
10192      0.0    2.7279
10193      0.2   -0.6048

```

[10194 rows x 21 columns]

In [17]:

```
# 3. Find the top 10 customers based on total sales.
```

```
top_customers = df.groupby('Customer Name')['Sales'].sum().sort_values(ascending=False).head(10)
```

```
print(top_customers)
```

stdout:

```

Customer Name
Sean Miller      25043.050
Tamara Chand     19052.218
Raymond Buch     15117.339
Tom Ashbrook     14595.620
Adrian Barton    14473.571
Ken Lonsdale     14175.229
Sanjit Chand     14142.334
Hunter Lopez     12873.298
Sanjit Engle     12209.438
Christopher Conant 12129.072
Name: Sales, dtype: float64

```

In [18]:

```
# 4. Calculate total profit by region and sort in descending order.
```

```
region_profit = df.groupby('Region')['Profit'].sum().sort_values(ascending=False)
```

```
print(region_profit)
```

stdout:

```

Region
West      110798.8170
East      94883.2603
South     46749.4303
Central   39865.3070
Name: Profit, dtype: float64

```

In [19]:

```
# 5. Create a pivot table showing sales by Category and Sub-Category.
```

```

pivot_table = pd.pivot_table(df,
                              values='Sales',
                              index='Category',
                              columns='Sub-Category',
                              aggfunc='sum')

```

```
print(pivot_table)
```

```

stdout:
Sub-Category    Accessories  Appliances      Art      Binders    Bookcases  \
Category
Furniture        NaN          NaN          NaN          NaN    115361.2043
Office Supplies  NaN    108213.185  27659.014  207354.881          NaN
Technology    167380.318          NaN          NaN          NaN          NaN

Sub-Category    Chairs    Copiers  Envelopes  Fasteners  Furnishings  \
Category
Furniture    335768.249          NaN          NaN          NaN    95598.126
Office Supplies  NaN          NaN    16528.362    8532.24          NaN
Technology      NaN    150745.29          NaN          NaN          NaN

Sub-Category    Labels    Machines      Paper    Phones    Storage  \
Category
Furniture        NaN          NaN          NaN          NaN          NaN
Office Supplies  12695.042          NaN    79540.538          NaN    224644.554
Technology        NaN    189925.031          NaN    331842.64          NaN

Sub-Category    Supplies      Tables
Category
Furniture        NaN    208020.182
Office Supplies  46725.498          NaN
Technology        NaN          NaN

```

```
In [20]:
```

```

# 6. Find the average discount given in each segment.

avg_discount = df.groupby('Segment')['Discount'].mean()

print(avg_discount)

```

```

stdout:
Segment
Consumer      0.157605
Corporate      0.156489
Home Office    0.147082
Name: Discount, dtype: float64

```

```
In [21]:
```

```

# 7. Identify the most profitable product in each category.

profitable_product = df.loc[df.groupby('Category')['Profit'].idxmax(),['Category', 'Product Name', 'Profit']]

print(profitable_product)

```

```

stdout:
Category      Product Name  \
3584  Furniture  Riverside Palais Royal Lawyers Bookcase, Royal...
6674  Office Supplies  GBC Ibimaster 500 Manual ProClick Binding System
5894  Technology      Canon imageCLASS 2200 Advanced Copier

Profit
3584  1013.127
6674  4946.370
5894  8399.976

```

```

In [24]:

# 8. Calculate month-wise sales trend using the Order Date column.

# Correctly parse date format (IMPORTANT FIX)
df['Order Date'] = pd.to_datetime(df['Order Date'], dayfirst=True)

# Extract Month-Year
df['Month'] = df['Order Date'].dt.to_period('M')

# Month-wise sales trend
monthly_sales = df.groupby('Month')['Sales'].sum()

print(monthly_sales)

```

```

stdout:
Month
2019-01    14518.0550
2019-02     4519.8920
2019-03    56933.9090
2019-04    28295.3450
2019-05    26319.7670
2019-06    34669.4796
2019-07    33946.3930
2019-08    28918.3385
2019-09    82670.4288
2019-10    32413.3390
2019-11    78826.9567
2019-12    72008.3085
2020-01    18461.9156
2020-02    11951.4110
2020-03    39978.6280
2020-04    34195.2085
2020-05    30246.9465
2020-06    24797.2920
2020-07    28765.3250
2020-08    37057.0042
2020-09    64627.0460
2020-10    31407.1335
2020-11    75972.5635
2020-12    75532.5572
2021-01    18830.3310
2021-02    22978.8150
2021-03    53031.0690
2021-04    38829.1590
2021-05    57042.8380
2021-06    40937.1480
2021-07    40300.4990
2021-08    31716.8143
2021-09    73521.6629
2021-10    59831.0010
2021-11    79411.9658
2021-12    97502.2770
2022-01    44259.2140
2022-02    20301.1334
2022-03    60728.4808
2022-04    36779.0361
2022-05    45155.4822
2022-06    53056.0777
2022-07    45989.4960
2022-08    64129.7580
2022-09    88064.5320
2022-10    83474.7832
2022-11    118454.5050
2022-12     85175.0328
Freq: M, Name: Sales, dtype: float64

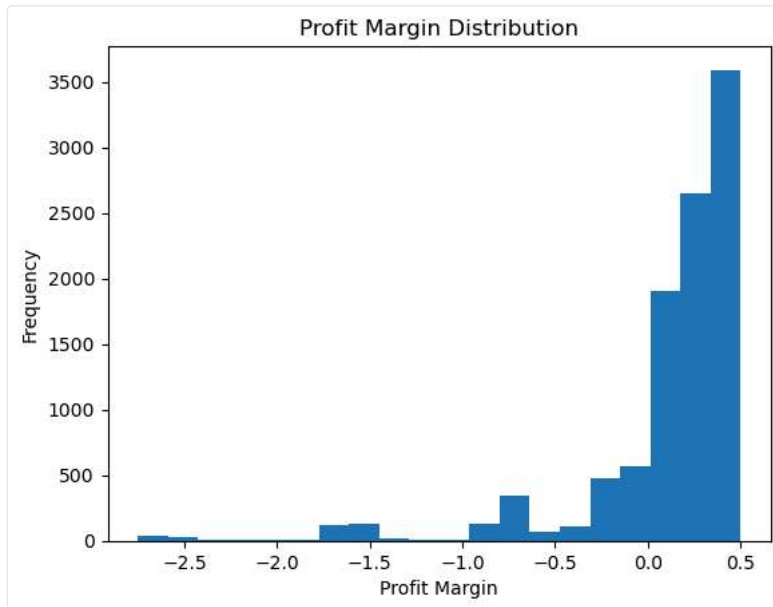
```

```
In [25]:  
  
# 9. Perform groupby operation to find total quantity sold per region.  
  
Region_Quantity = df.groupby('Region')['Quantity'].sum()  
  
print(Region_Quantity)
```

```
stdout:  
Region  
Central      8820  
East         11159  
South         6209  
West         12466  
Name: Quantity, dtype: int64
```

```
In [37]:  
  
# 10. Create a new column Profit Margin and analyze its distribution.  
import matplotlib.pyplot as plt  
  
df['Profit Margin'] = df['Profit'] / df['Sales']  
  
print(df['Profit Margin'].describe())  
  
df['Profit Margin'].plot(kind='hist', bins=20)  
  
plt.title("Profit Margin Distribution")  
plt.xlabel("Profit Margin")  
plt.ylabel("Frequency")  
plt.show()
```

```
stdout:
count    10194.000000
mean      0.122218
std       0.464890
min       -2.750000
25%       0.075000
50%       0.270000
75%       0.362500
max        0.500000
Name: Profit Margin, dtype: float64
```



```
In [42]:
# 11. Filter orders where sales are above 500 and profit is negative.

filtered_orders = df[(df['Sales'] > 500) & (df['Profit'] < 0)]

print(filtered_orders)
```

stdout:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	\
130	131	US-2019-113880	2019-03-01	05-03-2019	Standard Class	
145	146	US-2019-105648	2019-03-03	07-03-2019	Standard Class	
170	171	US-2019-140116	2019-03-10	17-03-2019	Standard Class	
198	199	US-2019-148838	2019-03-17	21-03-2019	Standard Class	
199	200	US-2019-148838	2019-03-17	21-03-2019	Standard Class	
...	...	...	...	...	...	
9943	9944	US-2022-168389	2022-12-11	17-12-2022	Standard Class	
9971	9972	US-2022-113992	2022-12-14	19-12-2022	Standard Class	
9972	9973	US-2022-151799	2022-12-14	18-12-2022	Standard Class	
9987	9988	CA-2022-151799	2022-12-14	18-12-2022	Standard Class	
10068	10069	US-2022-150910	2022-12-22	28-12-2022	Standard Class	

	Customer ID	Customer Name	Segment	Country/Region	City	\
130	VF-21715	Vicky Freymann	Home Office	United States	Elmhurst	
145	EJ-14155	Eva Jacobs	Consumer	United States	San Diego	
170	KT-16480	Kean Thornton	Consumer	United States	Denver	
198	CP-12340	Christine Phan	Corporate	United States	New York City	
199	CP-12340	Christine Phan	Corporate	United States	New York City	
...	...	...	...	...	...	
9943	DV-13045	Darrin Van Huff	Corporate	United States	Jacksonville	
9971	LC-16885	Lena Creighton	Consumer	United States	Plano	
9972	BF-11170	Ben Ferrer	Home Office	United States	Lawrence	
9987	BF-11170	Ben Ferrer	Home Office	Canada	Toronto	
10068	JL-15130	Jack Lebron	Consumer	United States	Jackson	

	Product ID	Category	Sub-Category	\
130	...	FUR-CH-10000863	Furniture Chairs	
145	...	FUR-TA-10002958	Furniture Tables	
170	...	OFF-ST-10000078	Office Supplies Storage	
198	...	FUR-TA-10001950	Furniture Tables	
199	...	FUR-TA-10003473	Furniture Tables	
...	...	...	...	
9943	...	FUR-TA-10004289	Furniture Tables	
9971	...	FUR-TA-10000577	Furniture Tables	
9972	...	FUR-TA-10003473	Furniture Tables	
9987	...	OFF-ST-10002790	Office Supplies Storage	
10068	...	FUR-TA-10002903	Furniture Tables	

	Product Name	Sales	Quantity	\
130	Novimex Swivel Fabric Task Chair	634.116	6	
145	Bevis Oval Conference Table, Walnut	626.352	3	
170	Tennsco 6- and 18-Compartment Lockers	636.408	3	
198	Balt Solid Wood Round Tables	1071.576	4	
199	Bretford Rectangular Conference Table Tops	1579.746	7	
...	...	...	...	
9943	BoxOffice By Design Rectangular and Half-Moon ...	721.875	6	
9971	Bretford CR4500 Series Slim Rectangular Table	974.988	4	
9972	Bretford Rectangular Conference Table Tops	526.582	2	
9987	Safco Industrial Shelving	526.582	2	
10068	Bevis Round Bullnose 29" High Table Top	934.956	6	

	Discount	Profit	Month	Profit Margin
130	0.30	-172.1172	2019-03	-0.271429
145	0.20	-23.4882	2019-03	-0.037500
170	0.20	-15.9102	2019-03	-0.025000
198	0.40	-553.6476	2019-03	-0.516667
199	0.40	-447.5947	2019-03	-0.283333
...	...	...	...	...
9943	0.45	-420.0000	2022-12	-0.581818
9971	0.30	-97.4988	2022-12	-0.100000
9972	0.30	-52.6582	2022-12	-0.100000
9987	0.30	-52.6582	2022-12	-0.100000
10068	0.40	-249.3216	2022-12	-0.266667

[263 rows x 23 columns]

```
In [44]:

# 12. Find duplicate records and remove them properly.

# Find duplicate records
print("Total duplicates:", df.duplicated().sum())

# Remove duplicate records
df_cleaned = df.drop_duplicates()

# Check new shape after removing duplicates
print("Shape after removing duplicates:", df_cleaned.shape)
```

```
stdout:
Total duplicates: 0
Shape after removing duplicates: (10194, 23)
```

```
In [45]:

# 13. Perform multi-index grouping using Region and Segment.

multi_group = df.groupby(['Region', 'Segment'])['Sales'].sum()

print(multi_group)
```

```
stdout:
Region  Segment
Central  Consumer      253962.2160
         Corporate    157995.8128
         Home Office    91212.6440
East     Consumer      357141.8610
         Corporate    203576.6250
         Home Office    131109.6820
South    Consumer      195580.9710
         Corporate    121885.9325
         Home Office    74255.0015
West     Consumer      363974.7390
         Corporate    232347.7645
         Home Office    143491.1050
Name: Sales, dtype: float64
```

```
In [47]:

# 14. Merge two datasets (simulate Orders and Returns dataset).

orders = pd.read_csv('Sample.csv')

# Simulate Returns dataset
returns = pd.DataFrame({
    'Order ID': ['CA-2019-100006', 'CA-2019-100090', 'CA-2019-100293'],
    'Returned': ['Yes', 'Yes', 'Yes']
})

# Merge datasets on Order ID
merged_df = pd.merge(orders, returns, on='Order ID', how='left')

print(merged_df.head())
```

stdout:

```
      Row ID      Order ID  Order Date  Ship Date      Ship Mode Customer ID \
0         1  US-2019-103800  03-01-2019  07-01-2019  Standard Class  DP-13000
1         2  US-2019-112326  04-01-2019  08-01-2019  Standard Class  PO-19195
2         3  US-2019-112326  04-01-2019  08-01-2019  Standard Class  PO-19195
3         4  US-2019-112326  04-01-2019  08-01-2019  Standard Class  PO-19195
4         5  US-2019-141817  05-01-2019  12-01-2019  Standard Class  MB-18085

      Customer Name      Segment Country/Region      City ... Region \
0  Darren Powers      Consumer  United States      Houston ... Central
1  Phillina Ober      Home Office  United States  Naperville ... Central
2  Phillina Ober      Home Office  United States  Naperville ... Central
3  Phillina Ober      Home Office  United States  Naperville ... Central
4    Mick Brown      Consumer  United States  Philadelphia ... East

      Product ID      Category Sub-Category \
0  OFF-PA-10000174  Office Supplies      Paper
1  OFF-BI-10004094  Office Supplies      Binders
2  OFF-LA-10003223  Office Supplies      Labels
3  OFF-ST-10002743  Office Supplies      Storage
4  OFF-AR-10003478  Office Supplies      Art

      Product Name      Sales Quantity \
0  Message Book, Wirebound, Four 5 1/2" X 4" Form... 16.448 2
1      GBC Standard Plastic Binding Systems Combs 3.540 2
2      Avery 508 11.784 3
3      SAFCO Boltless Steel Shelving 272.736 3
4  Avery Hi-Liter EverBold Pen Style Fluorescent ... 19.536 3

      Discount  Profit  Returned
0         0.2   5.5512      NaN
1         0.8  -5.4870      NaN
2         0.2   4.2717      NaN
3         0.2 -64.7748      NaN
4         0.2   4.8840      NaN

[5 rows x 22 columns]
```

In [48]:

```
# 15. Use apply() function to categorize orders into High, Medium, Low sales.

# Function to categorize sales
def sales_category(x):
    if x > 1000:
        return "High"
    elif x > 500:
        return "Medium"
    else:
        return "Low"

# Apply function to create new column
df['Sales Category'] = df['Sales'].apply(sales_category)

print(df[['Sales', 'Sales Category']].head())
```

stdout:

```
      Sales Sales Category
0    16.448           Low
1     3.540           Low
2    11.784           Low
3   272.736           Low
4    19.536           Low
```

In [50]:

```
# 16. Identify outliers in Sales using IQR method.

# Step 1: Calculate Q1 and Q3
Q1 = df['Sales'].quantile(0.25)
Q3 = df['Sales'].quantile(0.75)

# Step 2: Calculate IQR
IQR = Q3 - Q1

# Step 3: Define bounds
lower_bound = Q1 - 1.5 * IQR
upper_bound = Q3 + 1.5 * IQR

# Step 4: Identify outliers
outliers = df[(df['Sales'] < lower_bound) | (df['Sales'] > upper_bound)]

print(outliers)
```

stdout:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode \
5	6	US-2019-167199	2019-01-06	10-01-2019	Standard Class
8	9	US-2019-167199	2019-01-06	10-01-2019	Standard Class
12	13	US-2019-167199	2019-01-06	10-01-2019	Standard Class
22	23	US-2019-109232	2019-01-13	16-01-2019	Second Class
29	30	US-2019-162775	2019-01-13	15-01-2019	Second Class
...	...	...	...	...	...
10121	10122	US-2022-145219	2022-12-24	25-12-2022	First Class
10137	10138	US-2022-114055	2022-12-25	29-12-2022	Second Class
10149	10150	US-2022-121741	2022-12-26	26-12-2022	Same Day
10169	10170	US-2022-163979	2022-12-28	02-01-2023	Second Class
10172	10173	US-2022-158526	2022-12-29	01-01-2023	Second Class

	Customer ID	Customer Name	Segment	Country/Region \
5	ME-17320	Maria Etezadi	Home Office	United States
8	ME-17320	Maria Etezadi	Home Office	United States
12	ME-17320	Maria Etezadi	Home Office	United States
22	ND-18370	Natalie DeCherney	Consumer	United States
29	CS-12250	Chris Selesnick	Corporate	United States
...	...	...	...	...
10121	RM-19675	Robert Marley	Home Office	United States
10137	MH-18115	Mick Hernandez	Home Office	United States
10149	YC-21895	Yoseph Carroll	Corporate	United States
10169	KH-16690	Kristen Hastings	Corporate	United States
10172	KH-16360	Katherine Hughes	Consumer	United States

	City	...	Category	Sub-Category \
5	Henderson	...	Furniture	Chairs
8	Henderson	...	Office Supplies	Binders
12	Henderson	...	Technology	Phones
22	Mount Pleasant	...	Furniture	Chairs
29	Bossier City	...	Office Supplies	Storage
...	...	...	...	...
10121	Los Angeles	...	Technology	Copiers
10137	Huntsville	...	Office Supplies	Paper
10149	Fremont	...	Office Supplies	Storage
10169	San Francisco	...	Office Supplies	Storage
10172	Louisville	...	Furniture	Chairs

	Product Name	Sales	Quantity \
5	Global Deluxe High-Back Manager's Chair	2573.820	9
8	Ibico Hi-Tech Manual Binding System	609.980	2
12	Wireless Extenders zBoost YX545 SOHO Signal Bo...	755.960	4
22	Global Highback Leather Tilter in Burgundy	545.940	6
29	Fellowes Stor/Drawer Steel Plus Storage Drawers	572.580	6
...	...	...	...
10121	Hewlett Packard LaserJet 3310 Copier	2879.952	6
10137	Xerox 1915	629.100	6
10149	Tennsco Single-Tier Lockers	750.680	2
10169	Adjustable Depth Letter/Legal Cart	725.840	4
10172	DMI Arturo Collection Mission-style Design Woo...	1207.840	8

	Discount	Profit	Month	Profit Margin	Sales	Category
5	0.0	746.4078	2019-01	0.29		High
8	0.0	274.4910	2019-01	0.45		Medium
12	0.0	204.1092	2019-01	0.27		Medium
22	0.0	87.3504	2019-01	0.16		Medium
29	0.0	34.3548	2019-01	0.06		Medium
...	...	...	...	...		...
10121	0.2	1007.9832	2022-12	0.35		High
10137	0.0	301.9680	2022-12	0.48		Medium
10149	0.0	37.5340	2022-12	0.05		Medium
10169	0.0	210.4936	2022-12	0.29		Medium
10172	0.0	314.0384	2022-12	0.26		High

[1183 rows x 24 columns]

```

In [55]:

# 17. Rank products based on total sales within each category.

product_sales = df.groupby(['Category', 'Product Name'])['Sales'].sum().reset_index()

product_sales['Rank'] = product_sales.groupby('Category')['Sales'] \
    .rank(ascending=False, method='dense')

result = product_sales.sort_values(['Category', 'Rank'])

print(result)

```

```

stdout:

      Category      Product Name \
205  Furniture  HON 5400 Series Task Chairs for Big and Tall
320  Furniture  Riverside Palais Royal Lawyers Bookcase, Royal...
40   Furniture  Bretford Rectangular Conference Table Tops
200  Furniture  Global Troy Executive Leather Low-Back Tilter
174  Furniture  Global Deluxe High-Back Manager's Chair
...   ...
1659 Technology  Maxell CD-R Discs
1792 Technology  Sannysis Cute Owl Design Soft Skin Case Cover ...
1675 Technology  Memorex Mini Travel Drive 4 GB USB 2.0 Flash D...
1652 Technology  Maxell 4.7GB DVD+R 5/Pack
1717 Technology  PNY Rapid USB Car Charger - Black

      Sales  Rank
205  21870.5760  1.0
320  15610.9656  2.0
40   13565.5715  3.0
200  12975.3820  4.0
174  12325.7500  5.0
...   ...   ...
1659   15.7600  407.0
1792   13.8600  408.0
1675   13.6160  409.0
1652   11.6820  410.0
1717    9.5880  411.0

[1849 rows x 4 columns]

```

```

In [54]:

# 18. Calculate cumulative sales over time.

df['Order Date'] = pd.to_datetime(df['Order Date'], dayfirst=True)

df = df.sort_values('Order Date')

df['Cumulative Sales'] = df['Sales'].cumsum()

print(df[['Order Date', 'Sales', 'Cumulative Sales']].head(10))

```

```

stdout:
  Order Date    Sales  Cumulative Sales
0 2019-01-03    16.448             16.448
1 2019-01-04     3.540             19.988
2 2019-01-04    11.784             31.772
3 2019-01-04   272.736            304.508
4 2019-01-05    19.536            324.044
5 2019-01-06  2573.820           2897.864
6 2019-01-06     5.480           2903.344
7 2019-01-06    12.780           2916.124
8 2019-01-06   609.980           3526.104
9 2019-01-06    31.120           3557.224

```

```

In [56]:

# 19. Reshape data using melt() and pivot() functions.

# Reshape using melt
melted_df = pd.melt(df,
                    id_vars=['Category', 'Sub-Category'],
                    value_vars=['Sales', 'Profit'])

print(melted_df.head())

# Pivot back to wide format
pivot_df = melted_df.pivot_table(index=['Category', 'Sub-Category'],
                                 columns='variable',
                                 values='value',
                                 aggfunc='sum')

print(pivot_df)

```

```

stdout:
   Category Sub-Category variable  value
0  Office Supplies      Paper   Sales   16.448
1  Office Supplies    Binders   Sales    3.540
2  Office Supplies     Labels   Sales   11.784
3  Office Supplies    Storage   Sales  272.736
4  Office Supplies       Art    Sales   19.536
variable      Profit      Sales
Category      Sub-Category
Furniture
  Bookcases   -3632.0736  115361.2043
  Chairs      27223.5323  335768.2490
  Furnishings  13891.7430   95598.1260
  Tables      -17753.2061  208020.1820
Office Supplies
  Appliances   18329.4844  108213.1850
  Art          6653.1962   27659.0140
  Binders      31426.1003  207354.8810
  Envelopes    6988.0247   16528.3620
  Fasteners    2428.6358    8532.2400
  Labels       5572.7780   12695.0420
  Paper        34511.5070   79540.5380
  Storage      21285.1115  224644.5540
  Supplies    -1171.3945   46725.4980
Technology
  Accessories  41936.6357  167380.3180
  Copiers      56093.9365  150745.2900
  Machines     3461.9769  189925.0310
  Phones       45050.8265  331842.6400

```

```

In [57]:

# 20. Create a correlation matrix and interpret key relationships.

# Select only numeric columns
numeric_df = df.select_dtypes(include=['number'])

# Create correlation matrix
correlation_matrix = numeric_df.corr()

print(correlation_matrix)

```

stdout:

	Row ID	Sales	Quantity	Discount	Profit	\
Row ID	1.000000	-0.008958	0.000502	-0.001474	0.003022	
Sales	-0.008958	1.000000	0.198457	-0.027756	0.481460	
Quantity	0.000502	0.198457	1.000000	0.007475	0.066145	
Discount	-0.001474	-0.027756	0.007475	1.000000	-0.218882	
Profit	0.003022	0.481460	0.066145	-0.218882	1.000000	
Profit Margin	0.001198	0.002707	-0.003830	-0.864821	0.223224	
Cumulative Sales	0.999896	-0.008507	0.000490	-0.001078	0.003380	

	Profit Margin	Cumulative Sales
Row ID	0.001198	0.999896
Sales	0.002707	-0.008507
Quantity	-0.003830	0.000490
Discount	-0.864821	-0.001078
Profit	0.223224	0.003380
Profit Margin	1.000000	0.000834
Cumulative Sales	0.000834	1.000000